WEEK 1 TASK
STRATEGIC PLANNING AND DATA EXPLORATION IN LOGISTICS

Project Report: Demand Forecasting and Inventory Planning for a Small Retail Logistics Operation

1. Executive Summary

This report proposes a data science project for a small retail and e-commerce logistics operation that frequently faces two problems: popular products run out of stock, while slower-moving products occupy warehouse space and working capital. The project will use historical sales transactions to understand demand patterns, forecast near-term product demand, and support evidence-based replenishment decisions. The plan combines data cleaning, exploratory data analysis (EDA), regression/time-series forecasting, product clustering, and a simple inventory decision rule. The project is a planning proposal; no performance improvement is claimed before the data is analyzed.

2. Background Research and Logistics Scenario

Scenario: A small retailer ships products from a central warehouse to customers. The operations team currently decides how much stock to order using recent sales and staff experience. This can lead to stockouts during demand peaks and excess inventory during quiet periods. The proposed analysis will use transaction history to estimate demand by product and week, identify groups of products that need different replenishment policies, and provide a practical dashboard or report for inventory planners.

Public-data option: The UCI Machine Learning Repository provides the Online Retail II dataset, which contains transaction-level retail data such as invoice number, product code, quantity, invoice date, unit price, customer ID, and country. It is suitable for demonstrating demand analysis, although it represents an online retailer rather than the hypothetical clinic or logistics company in this scenario. Dataset source: Chen, D. (2012), Online Retail II, UCI Machine Learning Repository, https://doi.org/10.24432/C5CG6D. The dataset is listed at https://archive.ics.uci.edu/dataset/502/online+retail+ii.

A smaller alternative is UCI's Daily Demand Forecasting Orders dataset. It contains 60 days of observations from a Brazilian logistics company, with 12 predictive attributes and daily order volume as the target. It can be useful for a compact forecasting exercise, but its short time span limits how confidently seasonality can be estimated. Source: https://archive.ics.uci.edu/dataset/409/daily+demand+forecasting+orders.

Research rationale: Demand forecasting is a regression/time-series problem because the aim is to estimate a numerical quantity for a future period. Clustering can group products with similar sales volume or variability, helping planners apply different service levels or review frequencies. Inventory optimization can then translate forecasts into suggested reorder quantities while considering lead time and safety stock. These methods should be compared with simple baselines rather than assuming a complex model will always perform better.

3. Project Objectives

Explore historical order and product data to identify demand trends, high-volume items, low-volume items, and unusual records.

Build a baseline forecast for weekly product demand and compare it with a regression-based model.

Segment products into meaningful groups using demand volume and variability.

Define a transparent replenishment rule that uses forecast demand, supplier lead time, and available stock.

Recommend practical next steps for inventory planning based on measured results and known limitations.

4. Key Performance Indicators (KPIs)

The first phase should focus on forecast error and stockout rate. Inventory turnover and excess stock can be added if reliable inventory and cost records are available. KPI definitions must remain consistent across reporting periods.

5. Data Requirements and Research Plan

Required fields (ideal operational dataset):

Order or invoice date and time

Product/SKU identifier and product description

Quantity sold or ordered, including clear treatment of returns and cancellations

Unit price or product cost, if financial KPIs are required

Warehouse or delivery region, if available

Inventory on hand, supplier lead time, and replenishment history, if available

For the public-data demonstration, the UCI Online Retail II transaction file can supply dates, stock codes, quantities, and prices. It does not by itself provide a complete warehouse inventory ledger or supplier lead times. Therefore, stockout rate and reorder recommendations would require additional operational data or clearly labelled illustrative assumptions. The report must not present simulated inventory results as observed facts.

6. Proposed Data Science Methods

6.1 Data cleaning and preparation

Load the file with pandas; inspect column names and data types; parse invoice dates; check missing product identifiers and dates; identify duplicate rows; and investigate cancelled invoices or negative quantities. Returns should be handled separately or according to a documented business rule rather than silently treated as ordinary sales. Aggregate valid sales into daily or weekly demand by product.

6.2 Exploratory data analysis (EDA)

Calculate total units sold by product and by week, plot demand over time, inspect weekday and seasonal patterns, identify unusually large transactions, and compare demand variability across products. EDA helps determine whether a simple baseline is sufficient and whether additional features such as weekday, month, or holiday indicators may be useful.

6.3 Regression and demand forecasting

Predict the next week's product demand using features such as recent demand (lag values), rolling averages, weekday/month indicators, and product identity. Start with a naive forecast (next period equals the most recent period) or moving average, then compare it with a regression model such as Random Forest or HistGradientBoostingRegressor. Split data chronologically: train on earlier dates and test on later dates. Randomly shuffling time-series rows can leak future information into training and should be avoided.

6.4 Clustering

Use product-level features such as average weekly demand, standard deviation, and percentage of zero-demand weeks. Standardize the features and apply K-means clustering to explore groups such as high-volume stable products, high-volume variable products, and slow-moving products. Cluster labels are descriptive, not automatically correct business categories; inspect the cluster summaries before assigning replenishment policies.

6.5 Inventory decision support / optimization

If inventory and supplier lead-time data are available, calculate a reorder point as expected demand during lead time plus safety stock. A simple starting rule is: reorder point = average daily demand × lead-time days + safety stock. Safety stock should reflect demand variability and the desired service level. This is a decision-support rule, not a complete optimization model; a more advanced model could minimize ordering, holding, and stockout costs subject to supplier and warehouse constraints.

7. End-to-End Strategic Roadmap

Step 1 — Define the business question (Week 1): Confirm the products, planning horizon, current pain points, available data, and decision owner. Agree on KPI definitions.

Step 2 — Collect data: Download a public dataset or obtain authorized transaction and inventory exports. Record source, date range, fields, license, and limitations.

Step 3 — Audit and clean data: Check schema, missing values, duplicates, invalid quantities, cancellations, and date coverage. Document all exclusions and transformations.

Step 4 — Explore patterns: Create product rankings, weekly demand plots, summary statistics, and variability measures. Identify data gaps and candidate features.

Step 5 — Build a baseline: Create a naive or moving-average forecast and calculate MAE and WAPE on a chronological holdout period.

Step 6 — Train and compare models: Fit a regression model using past-only features. Compare it with the baseline using the same test dates and metrics.

Step 7 — Segment products: Cluster product demand profiles and inspect the resulting groups for business usefulness.

Step 8 — Translate results into actions: If inventory data exists, simulate reorder points and compare stockout/excess-stock outcomes with the current policy.

Step 9 — Communicate and monitor: Present findings, limitations, and recommendations. Monitor forecast error and inventory KPIs monthly and retrain only when justified.

8. Python Code Illustration

The following examples illustrate the proposed workflow. Column names may need adjustment to match the downloaded file. The code is a starting point and should be run and checked against the actual dataset before results are reported.

```
import pandas as pd
import numpy as np
# 1. Load and inspect the data
df = pd.read_excel("online_retail_II.xlsx")
print(df.head())
print(df.info())
print(df.isna().sum())
# 2. Basic cleaning (verify column names and business rules first)
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"], errors="coerce")
df = df.dropna(subset=["InvoiceDate", "StockCode", "Quantity"])
# Cancellation invoices may start with 'C'; keep them separate for review.
df["InvoiceNo"] = df["Invoice"].astype(str)  # adjust if the column is named Invoice
cancelled = df["InvoiceNo"].str.startswith("C")
sales = df.loc[~cancelled & (df["Quantity"] > 0)].copy()
# 3. Aggregate daily demand by product
daily = (
sales.groupby([pd.Grouper(key="InvoiceDate", freq="D"), "StockCode"])["Quantity"]
.sum()
.reset_index()
.rename(columns={"InvoiceDate": "date", "StockCode": "sku", "Quantity": "demand"})
)
print(daily.head())
# 4. Example: create lag features for one SKU
sku = daily[daily["sku"] == daily["sku"].iloc[0]].sort_values("date").copy()
sku["lag_1"] = sku["demand"].shift(1)
sku["lag_7"] = sku["demand"].shift(7)
sku["rolling_mean_7"] = sku["demand"].shift(1).rolling(7).mean()
sku = sku.dropna()
# 5. Chronological train/test split (no random shuffle)
split = int(len(sku) * 0.8)
train, test = sku.iloc[:split], sku.iloc[split:]
features = ["lag_1", "lag_7", "rolling_mean_7"]
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(train[features], train["demand"])
pred = model.predict(test[features])
print("Test MAE:", mean_absolute_error(test["demand"], pred))
```

```
# 6. Simple clustering of product demand profiles
profile = daily.groupby("sku")["demand"].agg(["mean", "std"]).fillna(0)
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
X = StandardScaler().fit_transform(profile)
cluster_model = KMeans(n_clusters=3, random_state=42, n_init=10)
profile["cluster"] = cluster_model.fit_predict(X)
print(profile.groupby("cluster")[["mean", "std"]].mean())
```

Implementation note: the public dataset may name the invoice field differently depending on the file/version (for example, Invoice or InvoiceNo). Check df.columns and update the code accordingly. For reliable forecasting, create a complete date-by-product panel so dates with no sales are represented as zero demand where appropriate; otherwise, lag features may mean the previous observed sale rather than the previous calendar day. For a final model, evaluate on multiple rolling time windows and compare against a baseline.

9. Risks, Assumptions, and Limitations

Historical sales are not always the same as true demand: a stockout can hide demand that could not be fulfilled.

Returns, cancellations, bulk purchases, and data-entry errors can distort sales totals unless handled consistently.

Public retail transaction data may not contain inventory on hand, supplier lead times, or actual delivery performance.

A model can perform well on historical data but poorly when demand patterns change; use time-based testing and monitoring.

Clustering depends on chosen features and scaling. Business users should validate whether the groups are actionable.

Do not claim savings, accuracy improvements, or reduced stockouts until these outcomes have been measured on appropriate test data.

10. Expected Outcomes and Conclusion

The expected deliverables are a cleaned analysis-ready dataset, descriptive charts, a baseline and regression forecast comparison, product demand clusters, KPI definitions, and a practical set of inventory-planning recommendations. The analysis should help logistics decision-makers understand which products have predictable or volatile demand and where further inventory data is needed. Its value will be assessed using held-out forecast errors and, where suitable inventory data exists, a simulation of stockout and excess-stock outcomes. The project supports more transparent, data-driven decisions, but recommendations must be validated by operational staff before they are used in purchasing or fulfilment.

11. References

Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D. Dataset page: https://archive.ics.uci.edu/dataset/502/online+retail+ii

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KPI	Definition	Why it matters
Forecast MAE	Mean absolute error: average absolute difference between actual and forecast demand.	Shows the typical size of forecast errors in units; lower is better.
Forecast WAPE	Sum of absolute errors divided by sum of actual demand, multiplied by 100%.	Gives an interpretable overall error percentage when total actual demand is non-zero.
Stockout rate	Number of product-periods with unmet demand divided by total product-periods, expressed as a percentage.	Tracks how often inventory is insufficient to meet demand.

KPI	Definition	Why it matters
Inventory turnover	Cost of goods sold divided by average inventory value over the same period.	Indicates how efficiently inventory is moving; interpret alongside service levels.
Excess/obsolete stock	Units or value above a defined target or with no sales over a chosen period.	Highlights working capital and storage space tied up in slow-moving products.